The current Mini Motorways model

Prepared for Mohammad | 28 October 2026 | Branch mohammad

It learned greater consistency on one seed, then plateaued.

This report explains the current learner and the latest completed experiment. The objective remains 1,000 completed trips on Los Angeles in OriginalReference. That goal has not been met.

Latest four-hour run	Measured outcome
Best observed game	73 trips; 461.4 simulated seconds
Completed games	4,549
Policy selections	2,271,492
Mean score: first / last 100 games	24.88 / 52.10
Held-out final sampled scores	3, 0, 0 on seeds 2000-2002
Training device	CPU; four independent games, one shared model

How to read this report

Pages 2-5 explain what it sees, what it can do, and how learning works. Pages 6-9 show results, transfer, the original baseline and throughput. Pages 10-12 cover fixes, decisions, unresolved limits and the exact input schema.

Source: archived seed-1000-parallel-02 summary/config/monitor files. This was a resumed run, not training from random weights. OriginalReference is a reconstruction, not a validated exact copy of the commercial game. No additional training was performed to produce this report.

1 / What the model can see

It reads structured numerical state, not the rendered game image.

Information	What enters the policy
Map and roads	Playable area, terrain, infrastructure categories, eight connection directions and pending demolition
Buildings	Locations/footprints, kinds, colors, entrances, demand, reservations and overcrowding timers
Traffic	Per-tile car count, maximum speed and maximum current waiting time
Motorways	Endpoints, other-end coordinates, link number and retirement status
Global state	Materials, time, week, trips, current upgrade offers and action-selection stage

The important missing information

It does not receive each individual car's color, destination or route. It is not explicitly told which houses have a working connection to a matching destination. It must infer that relationship from the map.

The policy has no recurrent memory and receives no stack of previous observations. Current waiting and overcrowding timers provide some accumulated state, but they are not a memory of past decisions.

Future spawns, planned building changes and random-generator state are withheld. Cells outside playable bounds are zeroed. Values are normalized and clipped; multiple entities sharing a tile lose detail. London-specific rail/train channels are absent.

Exact representation: 50 channels over 50 x 30 cells plus 42 global values. Current internal diagnostics are exposed; this is not claimed to be identical to a human player's visual information. Source: training/environment.py and its documented allowlist.

2 / Actions and the network

All ordinary tools remain available, subject to current resources and targets.

Action	Selection after choosing a tool
Wait / upgrade	None; command completes immediately
Road, bridge, light, roundabout, remove	One tile
Road stroke / house driveway	Start and end/direction tiles
Create motorway	Two endpoints
Edit motorway	Existing endpoint plus two replacements
Remove motorway	An active endpoint

The action vocabulary contains 1,513 IDs: 1,500 cells and 13 tools/upgrade choices. A command uses one to four policy selections. Intermediate selections advance no game time; a completed command advances 20 ticks, normally one simulated second. Rejected commands consume time too.

A compact actor-critic network

Stage	Shape / role
Map encoder	50 x 30 x 50 input; 16 then 24 convolution filters, 3 x 3 kernels, stride 2
Combine	Flatten 2,496 features + 42 globals; project to 128 shared features
Actor	64 hidden units; 1,513 action scores, filtered by the mask
Critic	64 hidden units; one estimate of discounted future trip reward

450,610 trainable parameters are numerical weights adjusted by optimization, not 450,610 rules or games. Masks remove locally unavailable choices; the host still decides final legality, including partial strokes. No path planner supplies routes. Sources: train.py, environment.py and original checkpoint inspection.

3 / How it learns

Maskable PPO = Proximal Policy Optimization with unavailable actions masked out.

Step	What happens
Observe	Encode the current city and available choices.
Choose	Sample an allowed action from the actor's probabilities.
Collect	Record observations, actions, rewards and critic predictions.
Estimate	Compare outcomes with predictions to estimate advantage.
Update	Adjust weights using PPO's clipped policy objective, value loss and entropy term.
Repeat	Use the updated shared policy for new interactions.

The reward has only one component

reward = trips after the command - trips before the command

No completed trip means zero reward, even for a useful road. Two completed trips mean +2. There is no construction penalty, demolition penalty, survival bonus or explicit negative failure reward. Failure matters indirectly because it ends future opportunities for trips.

Cars take time to deliver, so useful early decisions may receive credit much later. PPO and the critic estimate this connection; they do not automatically know what makes a correct road network. Intermediate endpoint choices also affect the selection-level discount.

Exploration versus deterministic playback

Training samples actions to explore. Deterministic evaluation chooses the highest-probability allowed action. Watching the best recorded game replays exact commands; it does not ask the final model to recreate that game.

PPO is on-policy: updates use recent rollouts, not a persistent replay dataset. A parameter update or low value loss alone does not establish better gameplay. The original report contains the schematic clipped probability-ratio objective and detailed glossary.

4 / Current training configuration

Parallel games collect experience for one policy, not four independently trained models.

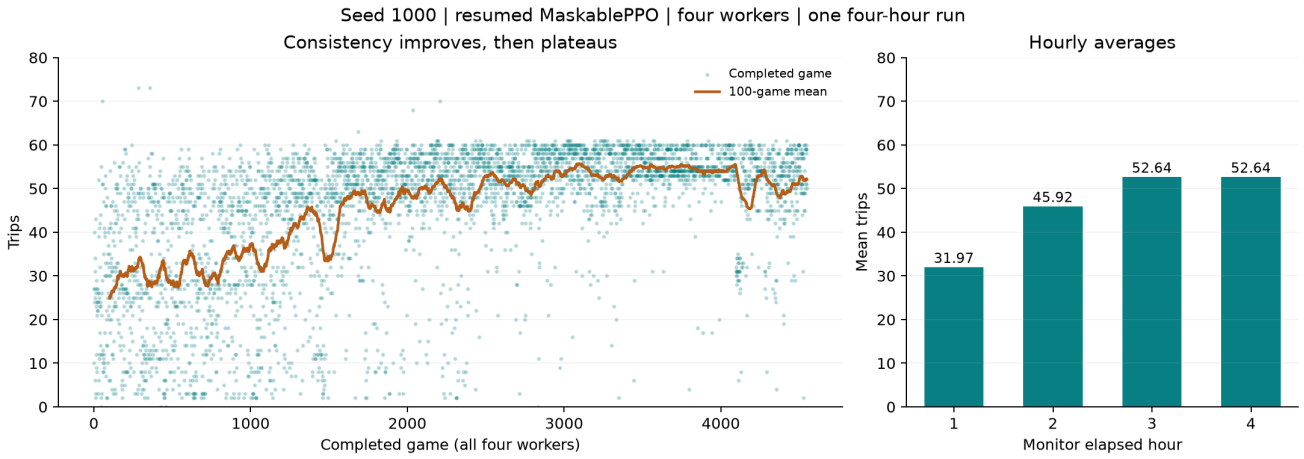
Setting	Current experiment
Version / city / materials	OriginalReference / Los Angeles / normal finite stock
Training seed	1000 only, repeatedly reset after game failure
Held-out seeds	2000, 2001, 2002; excluded from training
Workers / device	4 simulator hosts / CPU; two PyTorch threads
Aggregate rollout	256 selections = 64 per worker
Minibatch / epochs	64 / 4
Learning rate / optimizer	0.0003 / Adam
Gamma / GAE lambda	0.999 / 0.95
PPO clip / entropy coefficient	0.2 / 0.01
Value coefficient / gradient norm cap	0.5 / 0.5
Game cap	3,600 simulated seconds; ordinary failure usually occurs first
Latest learner budget	240 minutes or 10,000,000 selections, whichever first

The run stopped normally at its four-hour learner limit. Setup, initial evaluation and final evaluation are outside that timer. Resuming keeps policy/optimizer weights but starts new episodes, metrics and a fresh timestep counter; it is not a saved live city.

A label such as try 9/100 counts the current scheduling block. With one seed, reaching 100 starts another block on seed 1000; it does not stop training. Gamma discounts every policy selection, including intermediate selections. Shorter fragments with more workers may affect delayed credit. Sources: run config, train.py and parallel.py.

5 / Four hours: what improved?

The model improved its usual score on the practiced layout, without setting a new best.



Comparison	First 100 games	Last 100 games
Mean score	24.88	52.10
Games scoring 50 or more	7 / 100	75 / 100

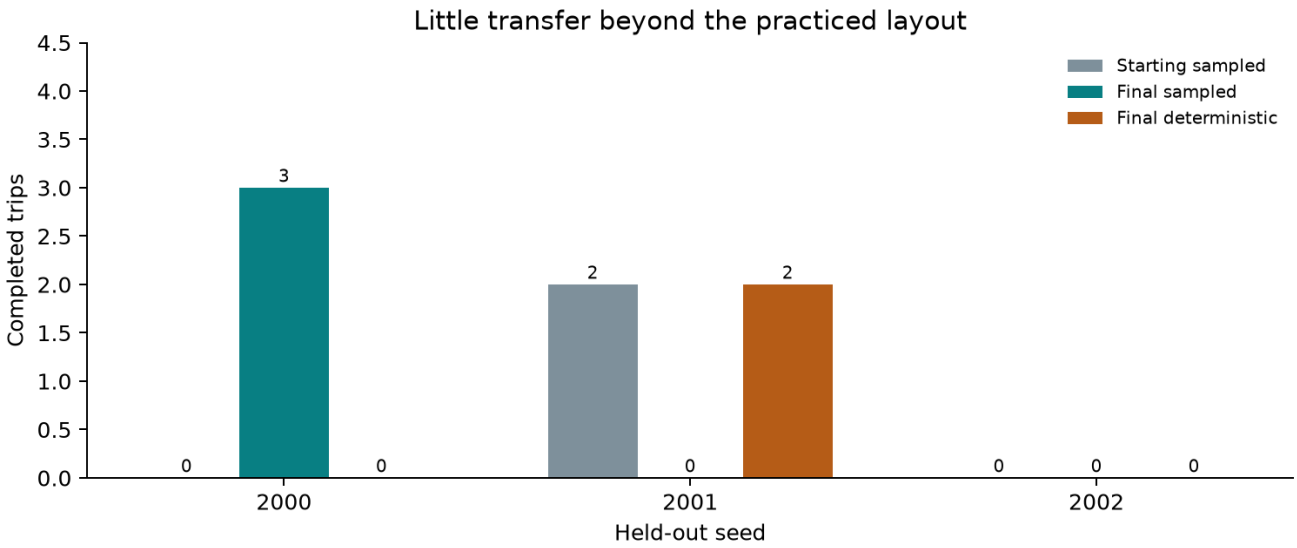
Hourly mean scores were 31.97, 45.92, 52.64 and 52.64. The final two hours show a plateau in these averages. The best game still scored 73, matching the earlier record.

Total / rate	Measured result
Learner wall time	14,400.7 seconds
Gameplay commands	1,650,223
Selections / second	157.7
Commands / second	114.6
Weight updates	Confirmed; final checkpoint saved

Data: archived worker monitor CSV and summary.json, 4,549 completed games. Games are merged by monitor completion time. The rolling mean covers 100 games. Hour bins use monitor time and may exclude boundary completions; setup precedes the learner timer. First/last comparisons are within this resumed run, not random initialization versus final policy.

6 / Did it learn other layouts?

The held-out evidence remains weak despite better training-seed consistency.



Seed	Starting sampled	Final sampled	Final deterministic
2000	0	3	0
2001	2	0	2
2002	0	0	0

On the practiced seed, the model learned a limited, repeatable pattern. On these unfamiliar seeds, final scores are only 0-3. This does not demonstrate a strong network-building strategy that transfers to other layouts.

Three held-out seeds are a small evaluation, not a reliable estimate across all city variations. A broader, predeclared test would be required for a general performance claim. No transfer to the commercial game has been tested.

Source: archived four-hour summary.json. Sampled and deterministic evaluations are different action-selection modes. The three evaluations are not independent training runs or a controlled comparison of learning algorithms.

7 / How this compares with the beginning

Historical baseline: the first run, not the current policy.



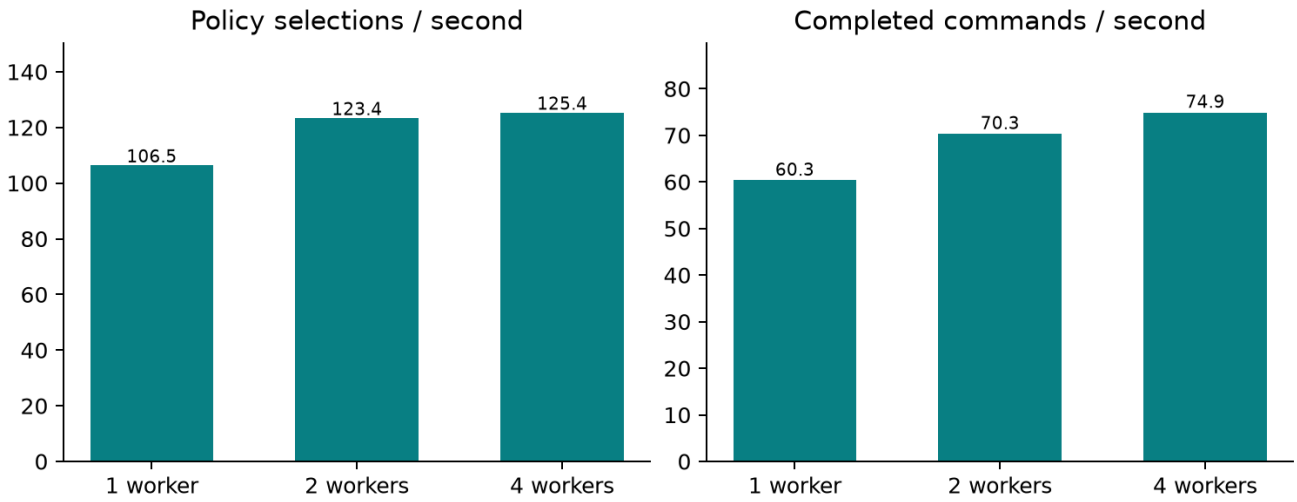
The first run completed 242 games in 431.1 seconds. Only four games earned any trips; the best was seven. All held-out evaluations scored zero. The final deterministic replay spent 25 road tiles, then waited 116 times and failed with no trips.

The later single-seed experiment clearly found more reliable ways to earn some trips. However, these runs differ in starting weights, seed schedule, duration and worker count. They do not isolate which change caused the improvement.

Baseline: first-full-tools, revision 30ef7ab, 65,536 selections, 35,610 commands, random seeds 1000-1099. Baseline mean score 0.0413; all 242 games failed normally. The original 16-page report is preserved unchanged, including survival, optimizer, material-use and network/parameter graphics.

8 / Can parallel games make it faster?

Yes for measured collection throughput; faster convergence is not established.



Two short trials at each worker count measured about 60.3, 70.3 and 74.9 commands/second with one, two and four workers. That is roughly 17% and 24% more commands with two and four workers than one in this benchmark.

These were 12-second trials after warm-up, in order 1/2/4/4/2/1, while another training run was active. They are under-load comparisons, not hardware maximums. Disposable clones performed brief learner updates without overwriting production weights.

The completed four-hour run averaged 114.6 commands/second under its own workload. That number is not directly comparable to the short under-load benchmark. Games/second also depends on how long the model survives: better play can make each game take longer.

Why speed does not scale fourfold

Simulator work runs in independent .NET processes, but Python JSON processing still shares the GIL. Policy calls are batched centrally. The thread-pool wrapper keeps large observation arrays in shared Python memory, avoiding another process-pipe copy.

Sources: archived parallel-benchmark.json, parallel.py and latest summary. No deliberate 0.8-games/second limit was configured. Throughput is not a prediction of time to 1,000 trips.

9 / Fixes since the first report

The learner, the viewer and the simulator need separate evidence.

Visible best-game recording

Training now saves exact ordinary commands for the best game globally and per seed. Ranking is highest trips, then longest survival, then fewer rejected actions. Partial games can qualify at stop. The native viewer supports play, step, 1x/4x/16x, restart and reload-best.

Cars appearing to drive across grass

Driveway edits removed the old connection from the drawn map immediately, while active cars retained committed lane snapshots and return routes. The viewer now draws pale gray retiring pavement until those trips clear. This does not reopen routing, alter stock or change the reward.

Ctrl+C crash during parallel training

The earlier shutdown waited on failed workers before saving, and invalid pipes caused another error. Windows hosts now have separate process groups; interruption saves final.zip atomically before worker cleanup. Worker/replay cleanup failures produce warnings after the checkpoint is safe.

The crashed run had no final.zip. Its valid latest.zip at 423,892 selections was used for the four-hour run. A periodic save can lose roughly the last two minutes before a crash. A new run folder is required when resuming.

What the checks establish

Forty simulator suites, seven environment tests, four workflow tests, three parallel tests, the shutdown test and five-map protocol checks passed at the fix revision. The earlier 73-trip/479.8-second/480-command replay verified in headless and native runtimes. That is a different recording from the latest 73-trip/461.4-second best.

Source: revision e6119eb and documented validation. Exact command replay and internal tests do not establish commercial-game parity. Roundabout connectors, parking and other rendering/geometry approximations remain documented limits.

10 / Decisions and what comes next

Preserve the task while testing specific explanations for the plateau.

Decision / change	Status
High score, OriginalReference, structured input, trial and error, all tools	User-selected task; 1,000 trips remains provisional goal
Branch mohammad; main stays playground	Implemented and published to GitHub
Repeated seed practice, then one seed only	Implemented; faster learning was a hypothesis, not a controlled result
Best-game maps and live seed/wait/ETA output	Implemented
Parallel collection and four-hour run	Implemented and explicitly authorized; run completed
Every update, decision, analysis and performance finding documented	Ongoing project requirement; report/graphs/evidence archived

Recommended next investigation, not implemented

Inspect recurring failure states and count working house-to-destination connections, material waste and missed demand. Then compare one targeted change at a time. Candidate bottlenecks include spatial action coordination, missing connectivity features, delayed credit, sparse reward and exploration.

No connectivity labels, new reward shaping, route suggestions, demonstrations or recurrent memory have been added. The observations suggest possible bottlenecks; they do not prove a single cause. More unchanged four-hour training is not currently recommended.

The original report's proposals are still proposals unless the history explicitly records their implementation. No new training was started for this report. User choices, engineering choices, measured facts and hypotheses are separated in training/EXPERIMENT_HISTORY.md.

11 / Exact inputs and evidence

Reference for checking what changed between model versions.

Grid channel	Encoding
0	Playable-cell visibility
1-3 / 4-8 / 9-11	Terrain / infrastructure / building-kind categories
12-19 / 20-27	Building colors / eight connection bits
28	Pending removal
29-31	Demand / 20; reservations / 20; overcrowding / 50
32-39	Eight entrance/driveway directions
40-42	Car count / 10; max speed / 2; max current wait / 60
43-47	Motorway endpoint; retirement; other x / 50, y / 30; number / 20
48-49	First selected cell; existing selected motorway endpoint

Global indices	Encoding
0-5	Six inventories: roads / 100, other materials / 10
6-8	Time / cap; week / 100; trips / 1000
9-24	Two offers: six type flags, amount / 100, roads / 100 each
25-28 / 29-41	Four action phases / thirteen selected tools

All inputs clip to [0, 1]. Schema: full-tools-staged-grid-v1. Unknown colors, changed grid sizes or more than two upgrade offers raise errors rather than being silently truncated.

Evidence and reproduction

Latest run: seed-1000-parallel-02; source e6119eb; resumed parallel-01/latest.zip; 240-minute budget. Sources: archived configs, summaries, episodes, monitor and optimizer CSV under docs/training/evidence-2026-10-28/. The evidence index includes hash manifests and all historical PDF figures. Checkpoints/player binaries remain local and are not backed up by the documentation archive.

Readable sources: training/README.md (contract), CURRENT_STATUS.md (latest results), EXPERIMENT_HISTORY.md (decisions/analysis), QUICKSTART.md (commands), docs/training/README.md (evidence). Rebuild source: docs/training/build_current_report.py. Algorithm explanations follow the implemented SB3-Contrib MaskablePPO setup and the original report's PPO references; no algorithm comparison was performed.